Compact physical Gaussian-ket models for homodyne quantum-state tomography

orangewk

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Abstract

Continuous-variable quantum-state tomography usually reconstructs a truncated Fock-basis density matrix. We study a compact alternative: finite mixtures of displaced, squeezed Gaussian kets whose $\rho = BB^\dagger$ construction is positive semidefinite by construction, fitted by per-sample homodyne likelihood with closed-form gradients, and composed with physical loss and noise channels. On the public propagating-light GKP dataset of Konno *et al.* (Science 2024), a rank-4 model with 92 real parameters matches the empirical full-rank maximum-likelihood frontier (255 parameters) at confidence-interval resolution on held-out likelihood. On a synthetic thermal-noise target that we prove lies outside the model family — no detection efficiency and no finite rank reproduces it exactly — the channel-composed model fitted blind exceeds a full-rank MLE run under the pre-declared 900-second baseline budget, a verdict that holds across all five pre-declared seed and noise configurations. We do not claim a universally superior method: comparisons on real data reuse observations across splits, the strongest baseline is test-selected, and the blind result covers one target class. The contribution is a compact, physically constrained model family together with a fully falsification-first research record — negative results, superseded scorings, and pre-declared protocols are all preserved in the accompanying repository.

1 Introduction

This project began with an analogy. In computer-vision 3D Gaussian splatting, a scene is represented as a mixture of anisotropic Gaussians and rendered by projection; in homodyne tomography, a quantum state's Wigner function is observed through one-dimensional projections — Radon transforms — indexed by the local-oscillator phase. A camera view corresponds to a homodyne phase, a splat to a phase-space Gaussian, rendering to a Radon projection. We state up front what the analogy is and is not: it is the origin of the project and an experimental representation choice, not a claim that computer-vision splatting solves quantum tomography.

Two model tracks came out of it. The first, *signed* anisotropic Gaussian mixtures fitted to quadrature marginals through a closed-form differentiable Radon forward model, is expressive and fast — signed weights can represent Wigner negativity — but a signed mixture is not guaranteed to be a quantum state, and the resulting overlap scores are not physically bounded and are not fidelities; on one out-of-family target the recorded score exceeds unity (§3.4). The second track responds to that tension: finite mixtures of displaced, squeezed Gaussian kets assembled as $\rho = BB^\dagger$, which is positive semidefinite by construction, admits a closed-form per-sample homodyne likelihood with analytic gradients, and composes with physical loss and noise channels. The physical track is the main subject of this paper.

Our contributions are deliberately scoped:

- **C1** — an honest scaling comparison of the signed-splat track against full-rank maximum-likelihood estimation on one, two, and three synthetic modes, including the non-PSD failure that motivates the physical track (§3.1).

- **C2** — on the public GKP homodyne dataset of Konno *et al.*, a rank-saturation analysis showing a 92-parameter physical model tying the empirical full-rank MLE frontier at confidence-interval resolution on held-out likelihood (§3.2).
- **C3** — a pre-declared blind gate on a synthetic thermal-noise target, together with an analytic proof that the target lies outside the model family for every assumed efficiency and every finite rank, and a five-configuration robustness sweep (§3.3).
- **C4** — a one-paragraph speculative discussion: the number of Gaussian components needed by a state as a possible continuous-variable analogue of stabilizer rank (§4).

Prior work bounds the novelty claims, and the closest precedents concern the physical track itself. Gaussian-superposition representations are an established lineage: the stellar representation of Chabaud, Markham, and Grosshans stratifies non-Gaussian states by the zeros of their Bargmann functions; Marshall and Anand simulate quantum optics by finite coherent-state decompositions; and variational Gaussian-wavepacket superpositions go back to Heller's frozen Gaussians in chemical physics. On the inverse-problem side, structured and model-based homodyne tomography is likewise established: Ohliger *et al.* formulated low-rank continuous-variable tomography, Tiunov *et al.* demonstrated that a physically constrained generative model beats generic MLE at small sample sizes on experimental homodyne data, and Fedotova *et al.* pursue Fock-truncation-free reconstruction at high amplitude. Against this background we do not claim a new Gaussian-state representation: the surviving contribution is algorithmic and empirical — a differentiable, closed-form-likelihood Gaussian-ket estimator with physical channel composition, evaluated under pre-declared falsification protocols. For the signed-splat track the anchors are differentiable Gaussian Radon tomography (R2-Gaussian, X^2-Gaussian) and Gaussian representations of Wigner negativity (Kenfack and Życzkowski; Tosca *et al.*), and the narrow open question is their combination on measured homodyne data; physical-model homodyne optimizers (Strandberg; Gaikwad *et al.*) are adjacent to both tracks. The repository's prior-art survey records these boundaries in detail.

2 Models and methods

2.1 Signed-splat Radon model

A state's Wigner function is modeled as a signed mixture of anisotropic Gaussians; the Radon projection of each component is Gaussian and closed-form, so quadrature histograms are differentiable in all mixture parameters. Gradient-driven birth, splitting, and pruning adapt the mixture size during fitting. We keep this section brief: in this paper the signed-splat model serves as the origin story and as one arm of the §3.1 scaling comparison, and its central limitation — signed mixtures need not be positive semidefinite, so overlap scores are not fidelities — is exactly what the physical track removes.

2.2 Physical Gaussian-ket mixtures

The physical model represents B as a K-component mixture of displaced, squeezed product kets and sets $\rho = BB^\dagger/\mathrm{Tr}(BB^\dagger)$: positive semidefinite by construction, with the trace normalized explicitly; a rank-R extension takes R such columns. The homodyne likelihood of a quadrature sample at a given phase is closed-form in the mixture parameters, with analytic gradients for the ket and mixture parameters (replacing an earlier finite-difference implementation, a 30–90× wall-clock improvement that made the multi-start protocols of §3.3 affordable); when the loss transmissivity η' is itself fitted, its derivative is taken by a scalar finite difference. Mixed states are reached by composing the ket mixture with a physical loss channel of transmissivity η'; the composed likelihood remains closed-form. The implementation also supports a *fixed* additive-noise variance, but the blind protocols of §3.3 deliberately do not use it: the fitted family is the loss-composed ket mixture alone, while the target contains additive thermal noise that this family provably cannot represent exactly — that mismatch is the point of the gate. The characteristic-function calculus behind the composition — including the commutation of loss and noise channels used throughout §3.3 — is developed in the repository's derivation document, and we state here only the facts the results depend on.

2.3 Baselines and scoring

The full-rank baseline is iterative $R\rho R$ maximum-likelihood estimation at a matched shot budget. Where wall-clock budgets matter (§3.3), the MLE runs under a pre-declared 900-second budget; the model side fits three initializations of roughly 470–670 s each — 1.4–2.0 $\times 10^3$ s aggregate per configuration — and selects blind by training likelihood. This asymmetry, and the fact that the MLE met its convergence criterion on only two of five sweep configurations, is disclosed wherever the comparison is cited.

Fidelity to a known synthetic target is scored with a generalized fidelity for subnormalized matrices, $F = \left(\mathrm{Tr}\sqrt{\sqrt{\rho}\,\sigma\sqrt{\rho}} + \sqrt{(1-\mathrm{Tr}\,\rho)(1-\mathrm{Tr}\,\sigma)}\right)^2$, which reduces to the Uhlmann fidelity at unit trace. The generalization matters because scored operators are crops of larger-cutoff computations: for an identical pair $\rho = \sigma$ of common trace $t < 1$, plain squared Uhlmann fidelity returns $t^2 < 1$ — its residual conflates crop loss with genuine disagreement — while the generalized form returns exactly 1. (The artifact was caught during review of the non-inclusion study.) All §3.3-class scores use the generalized form uniformly.

Held-out evaluation uses train/held-out splits with the selection rule declared before scoring. The known hazards of this design — a training likelihood nearly blind to large fidelity differences, and non-identifiable nuisance parameters — are documented negative results in this record (§3.4), and the blind protocols are shaped around them.

3 Results

3.1 Synthetic scaling: an honest one/two/three-mode story (C1)

The signed-splat track was compared against iterative $R\rho R$ maximum-likelihood estimation on simulated cat states at matched shot budgets. The outcome is deliberately reported as a scaling story rather than a win. At one mode the splat reaches a slightly higher score but the MLE is about twice as fast, so there is no computational advantage at that scale. At two modes, reconstruction fidelity is statistically indistinguishable across 20 paired seeds while the splat uses roughly 1/7.4 of the measured compute; the caveat is structural, since the result requires full cross-mode covariance and separable splats fail. At three modes a signed-splat run reached a higher Wigner-overlap score in about 15 s while the 512-dimensional MLE did not converge within 900 s — but the reconstruction is not positive semidefinite there, so this score is not a state fidelity and we do not report it as a physical-tomography win. That tension is what motivated the physical track.

3.2 Real GKP data: rank saturation and the frontier tie (C2)

On the public propagating-light GKP homodyne dataset of Konno *et al.* (Science 2024; Dryad doi:10.5061/dryad.t76hdr86j), a pure Gaussian-ket model initially lost clearly to full-rank MLE on held-out likelihood — a recorded negative result. Composing the model with a physical loss channel and moving to a rank-two squeezed-ket mixture improved held-out likelihood and, in a matched-degrees-of-freedom control, outperformed a rank-one model of comparable capacity.

A rank-saturation study then walked the remaining frontier gap down. The held-out rank curve saturates at $R = 4$–5; warm starts make material under-optimization unlikely under the tested schedule, and matched-degrees-of-freedom controls at two frontier points attribute each gain to rank rather than raw parameter count. At rank 4 — 92 real parameters — the physical model ties the empirical MLE frontier at confidence-interval resolution on both data reshuffles: conditional 95% intervals of $[-0.00002, +0.00020]$ and $[-0.00017, +0.00003]$ nats per held-out sample against the test-selected frontier best at 255 parameters. Three scope limits apply and are not footnotes: the reshuffled splits reuse the same observations, the MLE opponent is selected on test performance, and a tie at confidence-interval resolution is not a preregistered confirmation.

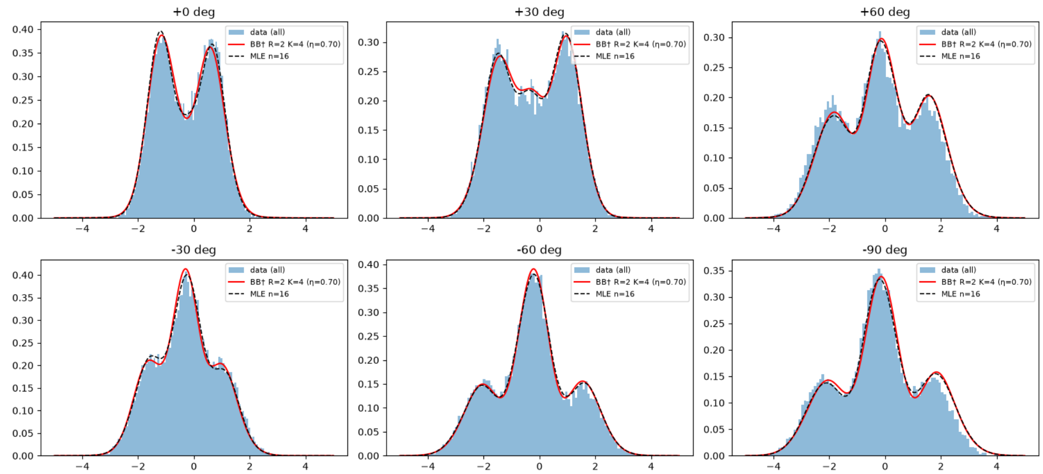

Figure 1: Measured GKP homodyne marginals and physical-model reconstructions.

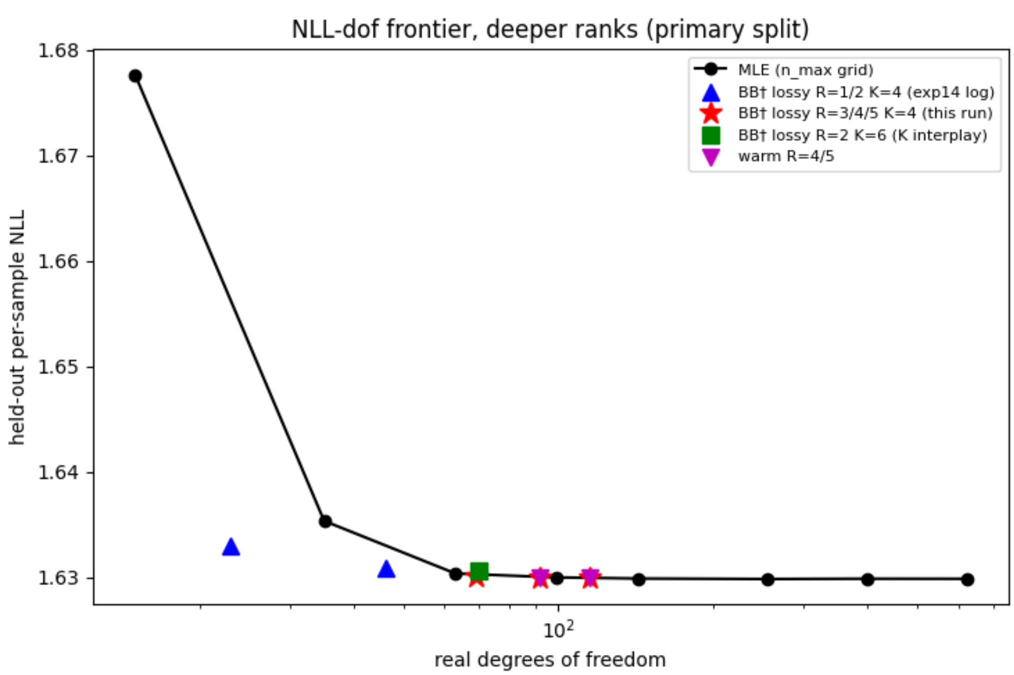

Figure 2: Held-out NLL versus degrees of freedom for physical models and MLE, ranks 1 through 5.

3.3 The blind gate and the family boundary (C3)

The strongest single result is a pre-declared, held-out gate on a synthetic target chosen to be hostile: a lossy cat state with added thermal noise, which is full-rank and — as proven below — outside the model family. Fitted blind (all modeling decisions declared before scoring), the loss-composed rank-2 model reached generalized fidelity 0.949 against 0.898 for a full-rank MLE run under the pre-declared 900-second baseline budget, with roughly 110 versus $\sim 2.6 \times 10^5$ real parameters. The pure-detection ket mixtures, by contrast, were capacity-limited: the rank-1 models landed within 2–3% of their rank-capacity ceiling (0.370–0.371 against 0.379), while the rank-2 mixture reached 0.648, about 86% of its 0.750 ceiling. The loss channel thus buys structured full-rank expressivity with very few parameters, rather than generic fitting capacity.

A non-inclusion analysis then settled what "out-of-family" means here. For **no** assumed detection efficiency $\eta' \in (0, 1]$ and **no** finite rank does the target factor as a loss-channel image of a finite-rank state: for η' below $\eta - \sigma$ the required pre-image is not a positive operator at all — positivity of the pre-image would force an s-ordered quasidistribution of the *cat state itself* to be nonnegative, and Gaussian smoothing of that nonnegative function would make the cat's Husimi function strictly positive, contradicting its exact (Bargmann) zeros, an adaptation of the phase-space argument of Lütkenhaus and Barnett; exactly at the boundary the pre-image is an amplified cat whose kernel has no finite-rank factorization; above the boundary the pre-image is a valid but full-rank state. Proofs are in the repository derivation; a validated numerical scan corroborates the theorems but is not the argument. The boundary is thin, however: direct best-approximation fits approach the target to $1\text{–}2 \times 10^{-3}$ in $1 - F$ (cutoff-stable best-found values, hence upper bounds on the true distance), with best-found fitted η' values of 0.648–0.661 across ranks — pressed against the positivity boundary (Fig. 3a). The blind gap of ~ 0.05 is therefore a fit- and data-budget effect, not the family boundary itself.

Finally, a robustness sweep repeated the blind comparison across three data seeds and a fourfold range of noise strength. The pre-declared verdict holds on all five configurations, with representative lossy fidelities 0.893–0.949 against MLE 0.815–0.936 (Fig. 3b). Budget disclosure: the MLE baseline runs under the pre-declared 900 s budget and met its convergence criterion on two of the five configurations; the model side fits three initializations ($1.4\text{–}2.0 \times 10^3$ s aggregate per configuration, sequential) and selects blind by training likelihood. Margins over the unconverged baselines are not guaranteed to survive longer MLE optimization. Three observations worth recording: no initialization-basin collapse occurred in any of the fifteen fits; the known selection hazard (§3.4) did appear in mild form — at the highest noise level the likelihood-selected initialization was the worst of three in fidelity, without affecting the verdict; and the fitted η' tracked the injected noise monotonically, exactly the flat-direction mechanism the non-inclusion derivation predicts.

3.4 Negative results and hazards (kept, not buried)

Two failure modes documented earlier in the record bound how far §3.3-style results can be trusted, and we state them as first-class findings. First, multi-seed refits of a lossy-target model exhibited a collapse basin: solutions differing by $\Delta F \approx 0.45$ in fidelity were separated by only $\approx 2.5 \times 10^{-3}$ nats per sample in training likelihood, and in one case a pre-declared selection rule picked a solution 0.04 worse in fidelity than a near-equivalent alternative at $\Delta\mathrm{NLL} \sim 10^{-4}$ nats per sample. Training likelihood can be almost blind to fidelity-relevant structure; every blind protocol above therefore pre-declares its selection rule, and the robustness sweep additionally records all per-initialization fidelities (the single-configuration gate retains only its selected fit — a limitation the sweep remedies). Second, jointly fitting the detection efficiency η with the state is non-identifiable in this design: fitted η scattered over 0.56–0.77 along a training-NLL plateau of width $\sim 10^{-5}$ nats per sample while fidelity varied from 0.06 to 0.80. Nuisance channel parameters must be measured or fixed, not fitted. Third, the signed-splat score is directly non-physical in the worst case: on a squeezed out-of-family target its recorded overlap score reached $1.7674 > 1$ — impossible for a physical state scored against a pure target — which is why §3.1 declines to report splat scores as fidelities. The earliest experiments in the repository predate the pre-declaration discipline and are recorded as exploratory.

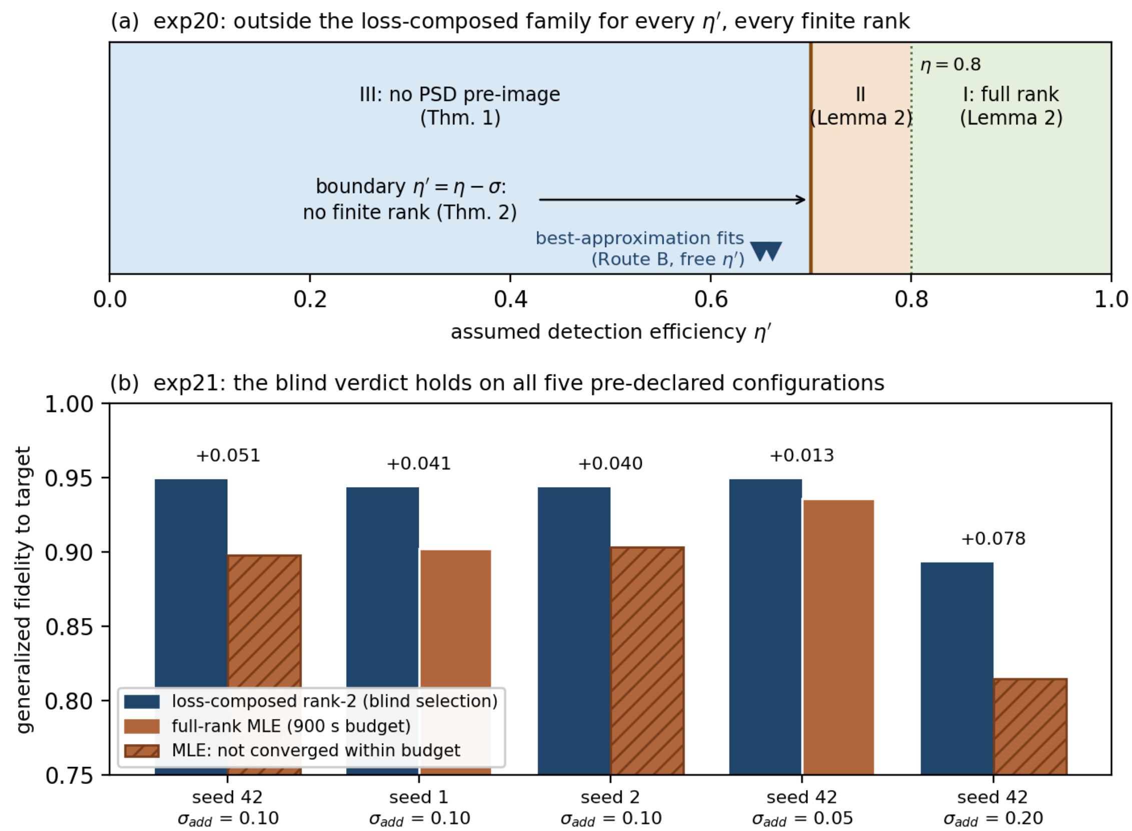

Figure 3: The blind gate and its family boundary. (a) The detection-efficiency axis with the excluding theorem for each regime; markers show best-approximation fits pressing against the positivity boundary. (b) The five-configuration robustness sweep; hatching marks MLE runs that did not meet their convergence criterion within the 900 s budget.

4 Discussion

What the record supports is compactness under physical constraints. On real data, 92 parameters tie a 255-parameter empirical frontier at confidence-interval resolution; on the synthetic gate, roughly 110 parameters exceed a $\sim 2.6 \times 10^5$-parameter baseline under the disclosed budget conventions. The controls isolate a mechanistic reading under the tested protocols: rank buys what parameter count alone does not (§3.2's matched-degrees-of-freedom controls), and channel composition buys structured full-rank expressivity that pure-detection mixtures lack (§3.3's ceiling analysis). The non-inclusion geometry sharpens this: best-approximation fits press against the positivity boundary of the channel parameter, with best-found fitted η' of 0.648–0.661 across ranks — the family is exploiting the boundary the theorems draw.

One speculative direction seems worth naming (C4). The number of Gaussian components a state requires — in either track — behaves like a resource count: cat states need few, and the rank/component saturation points found on real data (§3.2) are small. Hudson's theorem (pure states with nonnegative Wigner functions are Gaussian) anchors the intuition that non-Gaussianity is the resource being counted, suggesting a continuous-variable analogue of stabilizer rank: "how many Gaussians is this state," as a nonclassicality measure grounded in fitting practice. Any precise version must be distinguished from existing hierarchies: it is not the stellar rank of Chabaud *et al.* (a cat state has infinite stellar rank — its Bargmann function has infinitely many zeros — yet needs only two Gaussian kets here), and it is not identical to coherent-state decomposition rank (components here carry independent squeezing). Whether "Gaussian count" differs meaningfully from those hierarchies, and its basis dependence, robustness to loss, and relation to negativity measures, is exactly what would need to be established. We flag this as speculation: no result in this paper establishes it.

The limitations bound the claims. The blind gate covers one target class; the real-data analyses reuse observations across reshuffled splits and face a test-selected opponent; the wall-clock baselines are budget conventions, not converged optima; and nothing here is a universality claim about tomography methods.

5 Reproducibility

The repository (MIT license) and its archived Zenodo release (doi:10.5281/zenodo.21387212) contain the complete record. Experiment directories commit their scripts and available recorded outputs; the formal gates 16_exp11_seeds, 17_loss_control, 18_gkp_saturation, 19_thermal_gate, 20_noninclusion, and 21_thermal_sweep additionally commit machine-readable results.json files, and the summary figure regenerates from committed machine-readable result files (the sweep's results.json and the non-inclusion study's results_routeB.json) with assertions that fail if the plotted verdict diverges from the data. The research log preserves the chronological record, including superseded scorings and the negative results of §3.4. The public GKP dataset is included with its source attribution.

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Repository: https://github.com/orangewk/wigner-splat — prior-art survey, research log, derivations, and all experiment protocols.